Year 10 Foundation

REVISION PACK

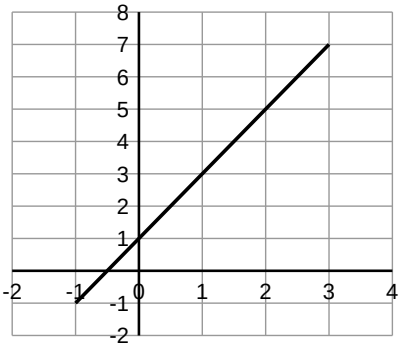
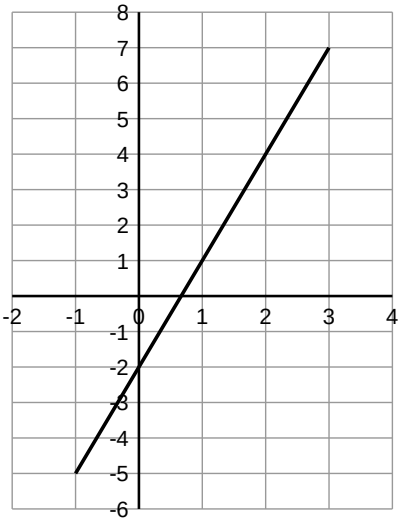
Graphs 1

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Graphs 1 · Gradients and straight-line graphs (Lessons 3.1, 3.2, 3.3)

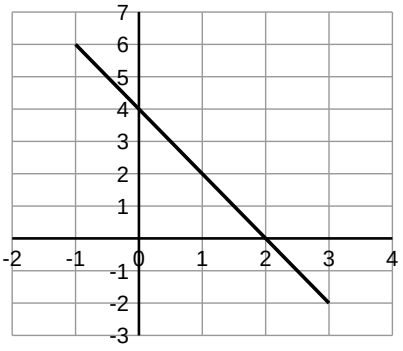
Q	Working	Answer
gradient from a graph		
1.1	$(0, -2)$ and $(2, 4)$ up 6, across 2 $\frac{6}{2} = 3$	3
1.2	$(-1, 5)$ and $(2, -4)$ up -9, across 3 $\frac{-9}{3} = -3$	-3
1.3	$(0, 1)$ and $(4, 3)$ up 2, across 4 $\frac{2}{4} = \frac{1}{2}$	$\frac{1}{2}$
1.4	$(-2, -4)$ and $(1, 5)$ up 9, across 3 $\frac{9}{3} = 3$	3
gradient from two points		
1.5	$\frac{18 - 3}{4 - 1} = \frac{15}{3}$ $= 5$	5
1.6	$\frac{1 - 10}{5 - 2} = \frac{-9}{3}$ $= -3$	-3
1.7	$\frac{6 - 2}{8 - 0} = \frac{4}{8}$ $= \frac{1}{2}$	$\frac{1}{2}$
1.8	$\frac{16 - 4}{3 - (-1)} = \frac{12}{4}$ $= 3$	3

1.9	$\frac{-1-7}{6-2} = \frac{-8}{4}$ $= -2$	-2
1.10	$\frac{10-(-2)}{5-1} = \frac{12}{4}$ $= 3$	3
drawing a straight-line graph		
1.11	$x = -1: \ y = 2 \times (-1) + 1 = -1$ <table><tr><td>x</td><td>−1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>y</td><td>−1</td><td>1</td><td>3</td><td>5</td><td>7</td></tr></table> 	**(a) y = −1, 1, 3, 5, 7** **(b) a straight line through the points**
1.12	$x = -1: \ y = 3 \times (-1) - 2 = -5$ <table><tr><td>x</td><td>−1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>y</td><td>−5</td><td>−2</td><td>1</td><td>4</td><td>7</td></tr></table> 	**(a) y = −5, −2, 1, 4, 7** **(b) a straight line through the points**

1.13

$x = -1 : \ y = -2 \times (-1) + 4 = 6$

x	−1	0	1	2	3
y	6	4	2	0	−2

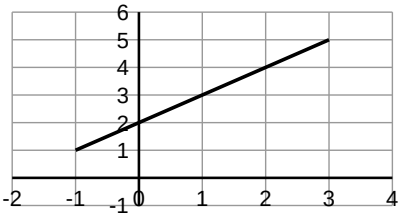


(a) y = 6, 4, 2, 0, −2
(b) a straight line through the points

1.14

$x = -1 : \ y = 1 \times (-1) + 2 = 1$

x	−1	0	1	2	3
y	1	2	3	4	5



(a) y = 1, 2, 3, 4, 5
(b) a straight line through the points

gradient and intercept

1.15

(a) $m = 4$

(b) $c = -3$

(a) 4
(b) (0, −3)

1.16

(a) $m = -5$

(b) $c = 8$

(a) −5
(b) (0, 8)

1.17

(a) $m = -1$

(b) $c = 6$

(a) −1
(b) (0, 6)

1.18

(a) $m = 3$

(b) $c = 2$

(a) 3
(b) (0, 2)

1.19	**(a)** $m = -4$ **(b)** $c = 9$	**(a) −4** **(b) (0, 9)**
1.20	**(a)** $m = 6$ **(b)** $c = -1$	**(a) 6** **(b) (0, −1)**

Graphs 1 · Parallel lines (Lesson 3.3)

Q	Working	Answer
parallel lines		
2.1	Same gradient, 2	$y = 2x + c$, **e.g.** $y = 2x + 4$ **(any** $c \neq 1$**)**
2.2	Same gradient, -4	$y = -4x + c$, **e.g.** $y = -4x$ **(any** $c \neq 3$**)**
2.3	Same gradient, $\frac{1}{2}$	$y = \frac{1}{2}x + c$ **(any** $c \neq -6$**)**
2.4	Both have gradient 5	$y = 5x + 1$ **and** $y = 5x - 3$
2.5	Gradient 3, y-intercept -1	$y = 3x - 1$
2.6	Gradient -2, y-intercept 4	$y = -2x + 4$